

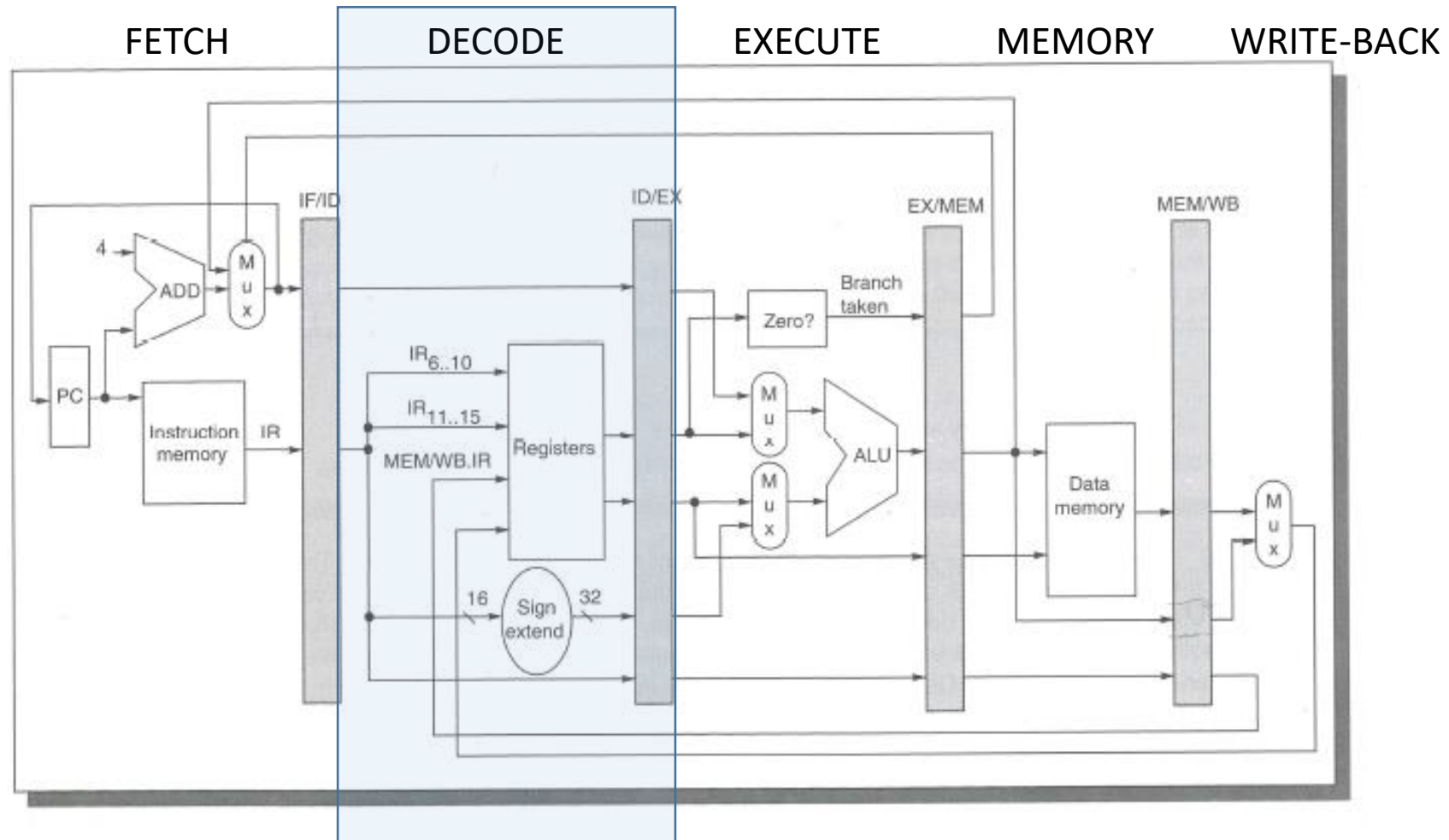
DLX Decode Stage

Dr. Jonathan Phillips

Utah State University

ECE 6740 – Advanced Reconfigurable Computing

DLX Pipeline Architecture



DLX Decode Stage

- Goal is to “parse” the instruction and prepare needed register and immediate values
 - Register Bank
 - Sign extender
 - Registered outputs
 - Logic to select instruction bits that address register bank??

Registers

- 32 registers, numbered R0 to R31

Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero

Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Do we need to initialize register values?

Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Do we need to initialize register values?
 - No! Program is in charge of setting and clearing registers.

Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Do we need to initialize register values?
 - No! Program is in charge of setting and clearing registers.
- We can think of the registers as a “small memory”
- Is it a RAM or a ROM?

Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Do we need to initialize register values?
 - No! Program is in charge of setting and clearing registers.
- We can think of the registers as a “small memory”
- Is it a RAM or a ROM?
- How many read ports do we need on the registers?

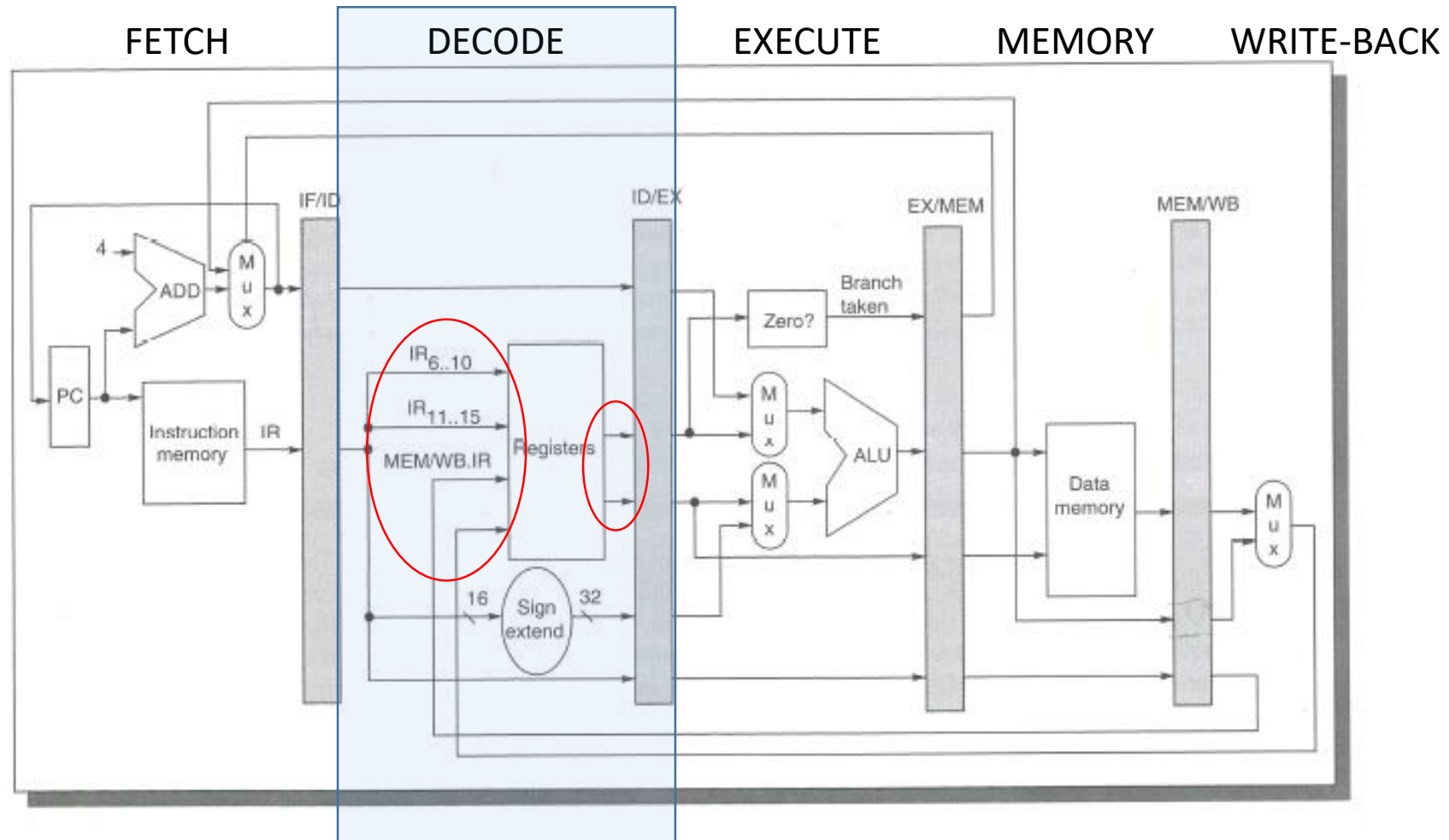
Registers

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Do we need to initialize register values?
 - No! Program is in charge of setting and clearing registers.
- We can think of the registers as a “small memory”
- Is it a RAM or a ROM?
- How many read ports do we need on the registers?
- How many write ports do we need on the registers?

VHDL RAM Example

- <https://www.intel.com/content/www/us/en/programmable/support/support-resources/design-examples/design-software/vhdl/vhd-single-port-ram.html>
- Chapter 11 of the textbook also has some good memory examples

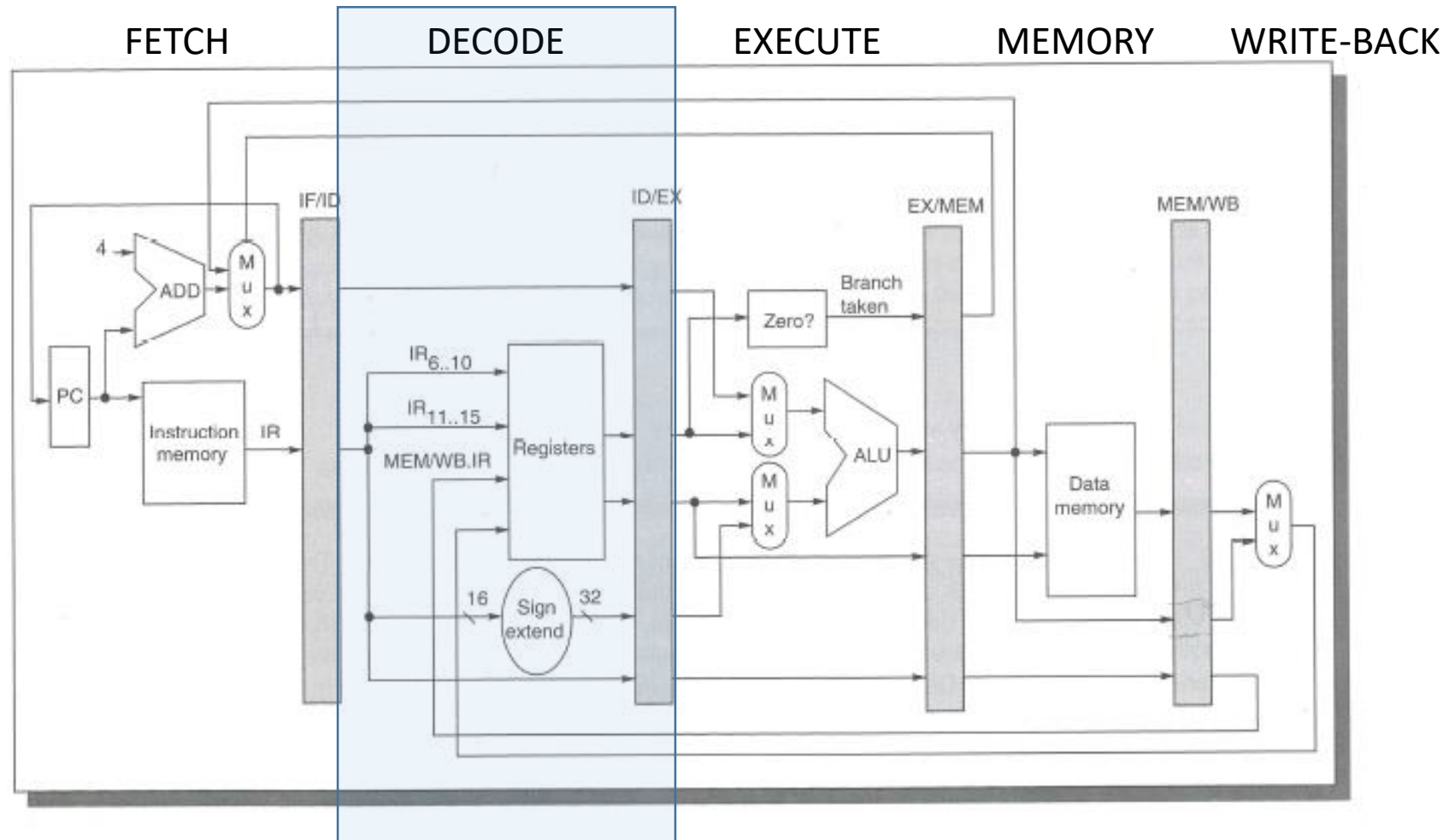
DLX Pipeline Architecture



Registers Summary

- 32 registers, numbered R0 to R31
- R0 **always** holds the value zero
- Think of a “Register File” with
 - 2 Read Ports – Retrieve data from source registers
 - 1 Write Port – Stick data in destination register
 - Write enable?

DLX Pipeline Architecture



Sign Extend

- Reasoning: Immediate values are 16 bits, we need to expand to 32

Sign Extend

- Reasoning: Immediate values are 16 bits, we need to expand to 32
- Cases to consider:
 - Immediate value is unsigned (how do we know?)

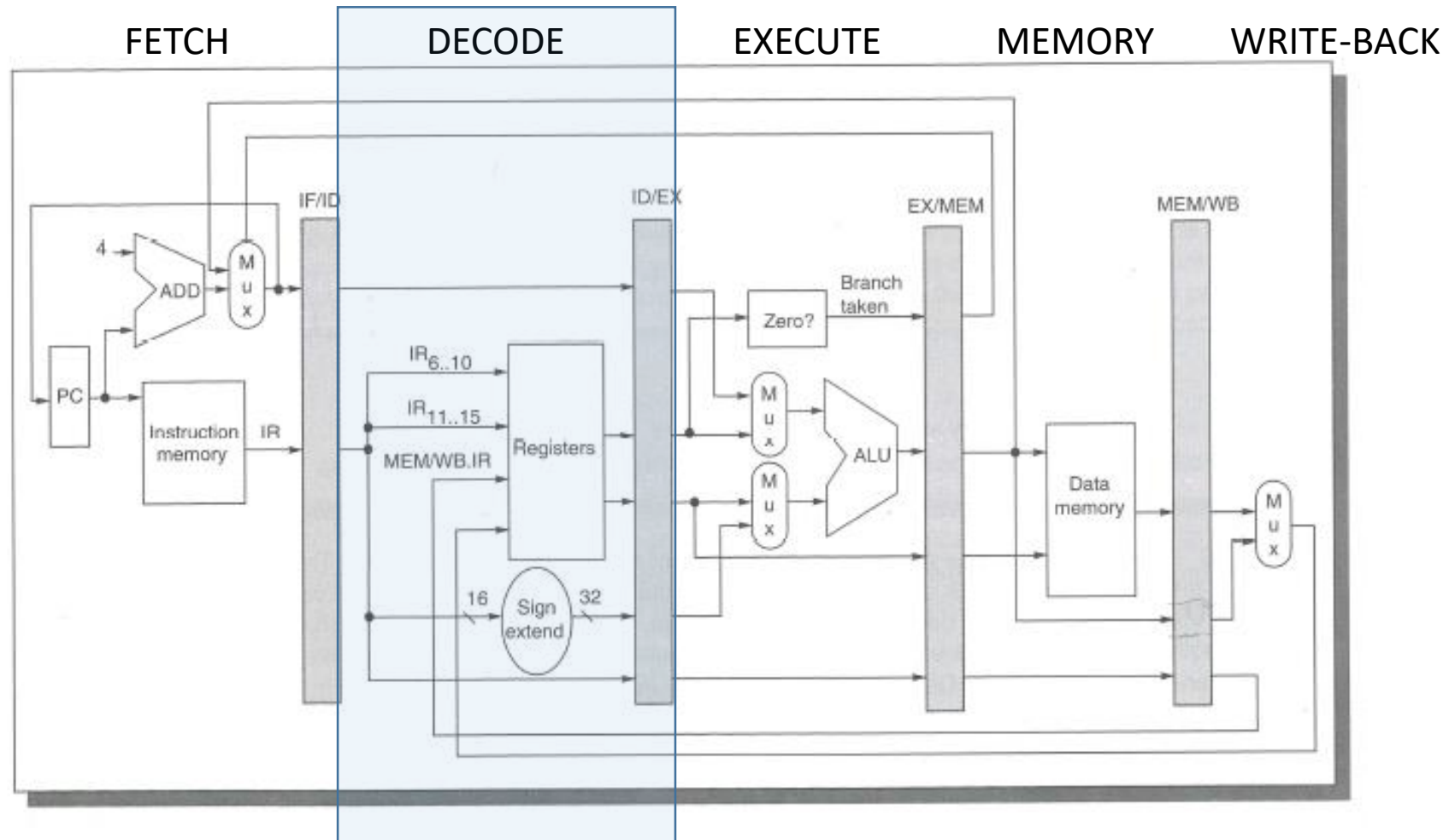
Sign Extend

- Reasoning: Immediate values are 16 bits, we need to expand to 32
- Cases to consider:
 - Immediate value is unsigned (how do we know?)
 - Immediate value is a signed positive number

Sign Extend

- Reasoning: Immediate values are 16 bits, we need to expand to 32
- Cases to consider:
 - Immediate value is unsigned (how do we know?)
 - Immediate value is a signed positive number
 - Immediate value is a signed negative number

DLX Pipeline Architecture



Thoughts and Ideas

- Decode Stage takes 1 clock cycle
- Remember that R0 must always be zero
- How are register read addresses determined?
- Could we have optimized our instruction format?